

$$X = \{x_i\}_{i=1..N} \quad \forall x_i \in \begin{matrix} \{H, T\} \\ \downarrow \downarrow \\ 1 \quad 0 \end{matrix}$$

□ Suppose they are i.i.d.

□  $\theta$  st  $P(X_i=H|\theta) = \theta$  ;  $E[X_i|\theta] = \theta$   
 $P(X_i=T|\theta) = 1-\theta$

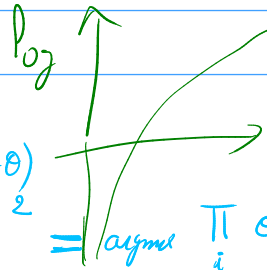
□ What is the MLE of  $\theta$   $\theta^{ML} = \underset{\theta}{\operatorname{argmax}} \mathcal{L}(\theta|X)$   
 maximum likelihood estimator

$N=3$

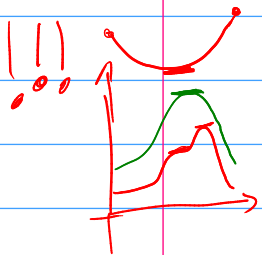
$$\underset{\theta}{\operatorname{argmax}} P(X|\theta) = \underset{\theta}{\operatorname{argmax}} P_{x_1, x_2, x_3}(\underset{H}{X_1=x_1}, \underset{T}{X_2=x_2}, \underset{T}{X_3=x_3} | \theta = \theta)$$

indep  $\Rightarrow P(A, B) = P(A)P(B)$   
 A and B

$$\theta^{ML} = \underset{\theta}{\operatorname{argmax}} \prod_i P(x_i|\theta) = \underset{\theta}{\operatorname{argmax}} \theta^{x_1} (1-\theta)^{1-x_1} \theta^{x_2} (1-\theta)^{1-x_2} \theta^{x_3} (1-\theta)^{1-x_3}$$



$$\begin{aligned}\theta^{ML} &= \operatorname{argmax}_{\theta} \log \left( \prod_i p(x_i | \theta) \right) &&= \operatorname{argmax} \log(\theta^1) + \log((1-\theta)^1) \\ & &&= \operatorname{argmax} 1 \times \log(\theta) + 2 \cdot \log(1-\theta) \\ \theta^{ML} &= \operatorname{argmax}_{\theta} \sum_{i=1}^N \log p(x_i | \theta) &&= \operatorname{argmax} \left( \sum_i x_i \right) \log(\theta) + (N - \sum_i x_i) \log(1-\theta)\end{aligned}$$



$$\operatorname{argmax}_{\theta} \underbrace{N_H \log(\theta) + N_T \log(1-\theta)}_{F(\theta)}$$

$$\boxed{\frac{\partial F(\theta)}{\partial \theta} = 0} \Rightarrow 0 = N_H \cdot \frac{1}{\theta} + N_T \cdot (-1) \frac{1}{1-\theta}$$

$$\Rightarrow 0 = N_H \frac{1-\theta}{(1-\theta)\theta} - N_T \cdot \frac{\theta}{(1-\theta)\theta}$$

$$\Rightarrow 0 = N_H (1-\theta) - N_T \theta$$

$$\Rightarrow \theta = \frac{N_H}{N_H + N_T} = \frac{1}{3} = \frac{3}{8}$$

$$P(X_{\text{new}} = H | \theta = \theta^{ML}) = \frac{N_H}{N}$$

$$P(X_{\text{new}} = T | \theta = \theta^{ML}) = \frac{N_T}{N}$$

$$X = \{x_i\}_{i=1..N}$$

$$\forall x_i \in \mathbb{R}$$

$$\mathcal{N}(\mu, \sigma^2)$$

variance ( $\sigma^2$ )

$\sigma$  = standard deviation

□ Suppose they are i.i.d.

$$\forall i \quad X_i \sim \mathcal{N}(\mu, 1^2)$$

$$\mathcal{N}(\mu, \sigma^2)(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}\right)$$

simulated from  
drawn from  
sampled from  
follows

MLE for  $\mu$  (in this case);  $\mu^{ML} = \arg\max_{\mu} p(X|\mu)$

$$\mu^{ML} \stackrel{iid}{=} \arg\max_{\mu}$$

$$\prod_i p(X_i|\mu) = \arg\max_{\mu} \prod_i \frac{1}{\sqrt{2\pi 1^2}} \exp\left(-\frac{1}{2} \frac{(x_i - \mu)^2}{1^2}\right)$$

$$= \arg\max_{\mu} \exp\left(\sum_i -\frac{1}{2 \cdot 1^2} (x_i - \mu)^2\right)$$

$\{\mu, \sigma\}$

$$\vec{\nabla} F(\vec{\theta}) = \vec{0}$$

$$F(\mu) = \arg\min_{\mu} \sum_i (x_i - \mu)^2$$

$$\frac{\partial F(\mu)}{\partial \mu} = 0 = \sum_i \frac{\partial (x_i - \mu)^2}{\partial \mu} = \sum_i (-1) \times 2 \times (x_i - \mu)$$

$$\Rightarrow \mu^{ML} = \frac{\sum_{i=1}^N x_i}{N}$$

$$= \sum_i \mu - \sum_i x_i$$

$$= N \cdot \mu - \sum x_i$$